Fiscal Year:	FY 2010	**Task Last Updated:**	FY 03/04/2010
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Project Title:	A Combinatorial Approach of Exercise and Myostatin Inhibition to Enhance Compromised Bone		
Division Name:	Human Research		
Program/Discipline:	NSBRI		
Program/Discipline--Element/Subdiscipline:	NSBRI--Musculoskeletal Alterations Team		
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Comments:			
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No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	1
No. of Bachelor's Candidates:	3	**Monitoring Center:**	NSBRI
Contact Monitor:		**Contact Phone:**	
Contact Email:			
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Key Personnel Changes/Previous PI:			
COI Name (Institution):	Phillips, Charlotte (MENTOR/University of Missouri-Columbia)		
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	POSTDOCTORAL FELLOWSHIP Osteogenesis imperfecta (OI) is a heritable connective tissue disorder caused by mutations in one of the two type I procollagen genes, COL1A1 and COL1A2. The resulting phenotype is short stature, bone deformity and numerous lifetime fractures. The oim/oim model mice are the most widely studied mouse model of OI. Oim/oim mice are homozygous for a null mutation in the COL1A2 gene of type I collagen and have significantly reduced femoral biomechanical strength and increased femoral stiffness as well as altered bone mineral composition. Heterozygous mice (+/oim) have a phenotype intermediate to oim/oim and wildtype (Wt) mice. G610C OI model mice represent a new mouse model of OI based on a large Amish population in which 64 individuals all carry the same COL1A2 mutation.		

Task Description:	Bone is inherently mechanosensitive, responding and adapting to its mechanical environment. Bone formation occurs in response to high mechanical loads, often changing its geometry to strengthen the skeleton. The largest physiological loads bones typically experience are from muscles with bone strength directly proportional to muscle mass. Myostatin (mstn) is a member of the transforming growth factor-ß (TGF-ß) super family and is a negative regulator of skeletal muscle growth. When the myostatin protein is missing or non-functional, the result is increased muscle growth with a concomitant increase in bone strength. Recently a completely myostatin deficient child was described whose quadriceps muscles were 7.2 standard deviations above normal but without any detrimental health consequences. Myostatin deficient mice have a similar phenotype: increased muscle mass and bone strength compared to Wt mice. Previous studies demonstrated that mice completely deficient for myostatin (mstn/mstn) have a 40% increase quadriceps muscle mass, an 11% increase in bone mineral density, and a 20% increase in radial ultimate force relative to Wt mice. When mstn/mstn mice were subject to treadmill exercise, radial ultimate force improved by an additional 17%. Exercise has also been shown to be beneficial for human patients at an increased risk for fracture (i.e. osteoporosis, osteogenesis imperfecta). Based on these previous reports, this project was designed to determine if a combination of exercise and myostatin haploinsufficiency could improve the bone strength in mice with compromised bone (+/oim and G610C OI). The aims of this project are threefold: 1) Determine if +/oim and G610C OI mice are able to tolerate and respond to treadmill exercise, 2) Determine if myostatin haploinsufficiency will ameliorate the bone phenotype of +/oim and G610C OI mice and 3) Determine if a combination of exercise coupled with myostatin haploinsufficiency will increase muscle mass and bone strength in +/oim and G610C OI mice. The long term goal of this project is to determine if a combination of these two therapies may be additive, further improving the bone phenotype beyond either therapy alone. This combination would be represent a novel treatment approach for OI patients, astronauts returning from microgravity, bed rest patients and others with compromised bone. Preliminary data demonstrates the reduction in bone strength seen in female +/oim animals was partially rescued when those mice either underwent treadmill exercise or were also haploinsufficient for the myostatin protein (+/mstn). Treadmill exercise did not appear to impact femoral geometry. However, modest improvements were seen in femoral torsional ultimate strength and energy to failure in both Wt and +/oim mice, though these improvements were not significant. Treadmill exercise did significantly reduce whole bone and material stiffness in both Wt and +/oim mice. Curiously, treadmill exercise did not appear to be beneficial for male Wt or +/oim mice. Myostatin haploinsufficiency marginally improved femoral torsional ultimate strength and energy to failure in +/oim mice, though not significantly. Myostatin haploinsufficiency did significantly reduce whole bone and material stiffness in +/oim mice. Taken together, these data indicate that both treadmill exercise and myostatin deficiency have the potential to improve bone strength and reduce stiffness in +/oim mice with compromised bone. In order to complete this study, additional female +/oim mice will be evaluated to assess the impact of myostatin haploinsufficiency on both muscle size and bone strength and stiffness. Additionally, the same studies will be performed on male +/oim mice to determine if a gender difference exists in the ability of the bone to respond to myostatin haploinsufficiency. Male and female G610C OI mice will also be evaluated for their ability to respond to treadmill exercise. The potential impact of maternal myostatin haploinsufficiency on the muscle size and bone strength of the offspring will also be evaluated. Finally, +/oim mice of both genders that are also haploinsufficient for myostatin will be subjected to treadmill exercise to determine if exercise coupled with myostatin haploinsufficiency further improves bone strength in +/oim mice.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Muscle atrophy and bone loss are both been associated with the weightlessness experienced by astronauts, but these are also widespread health concerns on Earth. As a person ages, muscle use tends to decrease. With disuse, muscle atrophy increases and bone strength decreases with a concomitant increase in fracture risk. While it has long been known that exercise decreases muscle atrophy and fracture risk while increasing bone strength, the role of myostatin haploinsufficiency may play in further augmenting the positive physiological effects of exercise is not known. This study is designed to determine 1) the effects of exercise on the bone strength in both normal mice and mice with compromised bone whose clinical outcome mimics that seen in osteoporosis patients, 2) what impact, if any, myostatin haploinsufficiency has on bone strength, muscle physiology and muscle strength in both normal mice and mice with compromised bone and 3) determine if a combination of exercise and myostatin haploinsufficiency will prevent both muscle atrophy and bone loss. At the completion of this study we will have determined whether a pharmacological approach of myostatin inhibition should be pursued, along with the inclusion of exercise, to potentially ameliorate the muscle atrophy and bone loss often seen as a consequence of a sedentary lifestyle.
Task Progress:	Toward Aim 1, we have completed characterization of female and male +/oim mice in response to treadmill exercise. The G610C OI mice are currently on the treadmill and should be ready for analyses in the coming months. Toward Aim 2, we are currently analyzing femurs collected from female and male +/oim mice that were also haploinsufficient for the myostatin protein to determine if reduction of myostatin protein will help ameliorate the bone phenotype of +/oim mice. Additional mice are needed to complete this study and those mice will be ready for analyses in the next few months. Following completion of these two aims, we will proceed to aims 4 and 5 and begin breeding the G610C OI model to mice haploinsufficient for the myostatin protein and subjecting these mice, as well as +/oim mice also haploinsufficient for myostatin, to treadmill exercise. Lastly, toward Aim 3, we are currently analyzing femora collected from Wt, +/oim, +/mstn and +/mstn +/oim mice born to mothers which were also haploinsufficient for the myostatin protein. The data from these femora is being compared to data from femora collected from Wt, +/oim, +/mstn and +/mstn +/oim mice born to mothers which were not haploinsufficient for the myostatin protein to determine if a maternal myostatin allele status impacts the muscle mass and bone strength in the offspring as has been reported in the literature.
Bibliography Type:	Description: (Last Updated: 08/24/2020)

Abstracts for Journals and Proceedings	Carleton SM, Weber BA, Ferreira JA, Brown M, Phillips CL. "Maternal myostatin deficiency reduces femoral bone strength in +/oim offspring with partial rescue in +/mstn +/oim offspring." 31st Annual Meeting of the American Society of Bone and Mineral Research, Denver, Colo., September 11-15, 2009. J Bone Miner Res. 2009:24(Suppl 1). , Sep-2009
Articles in Peer-reviewed Journals	Carleton SM, Whitford GM, Phillips CL. "Dietary fluoride restriction does not alter femoral biomechanical strength in col1a2-deficient (oim) mice with type I collagen glomerulopathy." Journal of Nutrition. 2010 Oct;140(10):1752-6. Epub 2010 Aug 19. PMID: 20724489 , Oct-2010